



(Don't Write anything in this Area)

- Q.3) (a) Secular trend in human growth can be positive, negative or neutral. Illustrate with examples. (20 Marks)

Secular trend in human growth refers to the trend which the human growth trajectory follows across the generation, due to bioplasticity of the gene (Eric Turkheimer)

Secular Trend as Positive

It means that growth pattern of population is increasing, consistent and providing long term benefits in Human characteristics across the generation.

It can be seen in $\Rightarrow$

- i) Height $\rightarrow$ is increasing in the subsequent generation
- ii) Weight $\Rightarrow$ is also increasing in subsequent generation.

iii) Age of Menarche $\Rightarrow$ is decreasing in the generation due to better nutrition and socio-cultural impact.

Examples supporting Positive Trend

(i) In the Japan, during 1950 - Average height was 160cm but during 1975 $\rightarrow$ it was 175cm, it was growing at 4cm/decade rate initially but later it came to 1cm/decade.

(ii) Similarly in Sweden & Norway, the decadal height growth rate during 1952-1982 was 0.3cm/decade.

(iii) AM Shah studied the Menarchial age of Gujarati girl was 14.8 and Delhi girl was 11.2 $\Rightarrow$ due to more stress $\rightarrow$ $\uparrow$ cortisol $\rightarrow$ $\uparrow$ sugar $\rightarrow$ $\uparrow$ fat
 (f) Better Nutrition $\rightarrow$ early 17/ fat.

(iv) It was also seen that on an average in Maharashtra, the younger generation is getting menarche at rate of 6 month/decade.

Negative Secular Trend

It is the pattern of growth wherein, there is constant, decline in the Human characteristics across the generation.

Reason → Biological factor- like decline in the organ functioning
 → Genes are not getting fully expressed.
 → Environmental factor- pollution, PM-2.5, PM 10, alcoholism, smoking etc

Examples →

① In the Gautemala, when there was intense civil war during

1972-1980's → there was decline in the average height of the people.

② As per NFHS-5, due to low per capita income, the Indian height is declining at rate of 6cm/decade.

Neutral secular trend

As per Malina, it means that gene has fully expressed itself in the given environment or it has poorly expressed itself in the environment.

③ In the Norwegian Country, the height has reached the Maximum so, there is less positive trend.

Similarly, in African war ridden countries, due to malnutrition, the growth has constantly declined to its lowest.

Hence we can see secular trend is positive, negative & Neutral.